method that matches the arguments and return type is executed. For example, if the method is called using two integers as an argument and returns an integer value, then the first method is executed. Similarly, the other two methods are executed as a result of the corresponding argument and return type. This phenomenon is known as method overriding. The same name can be used for methods in the parent and derived class.

1.5 Applications Of OOP

In terms of benefits, OOP offers various benefits to both the designer and the user of the program. The various benefits of OOP are as follows:

- Through the process of inheritance, redundant codes can be eliminated and the use of classes can be extended.

- Programs can be built from standard working modules that communicate with each other. This saves development time and increases productivity.

- Data hiding helps the programmer to build secure programs that cannot be touched by the code of other components of the program.

- OOP allows multiple instances of an object to co-exist without any interference.

- Mapping objects in the problem domain is also possible by OOP.

- The data-centred design approach in OOP captures more detail of a model.

- It also helps in proper communication between objects by various message passing techniques, simplifying the interface description.